Solution Requirements Specification (SRS)

Project: NeuroClass — Multimodal Attendance & Engagement Agent

Module: `2.Requirement Analysis`

Phase: Requirements Engineering (Functional & Non-Functional)

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1. System Overview & Scope

NeuroClass is a 100% offline edge classroom intelligence system engineered to automate attendance via sustained presence tracking, quantify multimodal classroom engagement through non-invasive computer vision and acoustic metrics, and provide real-time pedagogical nudges to instructors.

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2. Functional Requirements (FR) Traceability Matrix

The functional requirements map directly to the 6 system epics and 17 core engineering activities:

FUNCTIONAL REQUIREMENTS MATRIX			
Epic ID	Requirement ID	Requirement Description	System Module
EPIC 1	FR-1.1 (TASK-01)	Modular Architecture & System Configuration	backend.config
	FR-1.2 (TASK-02)	Edge Privacy & Local Data Persistence Schema	backend.db
EPIC 2	FR-2.1 (TASK-03)	Real-Time Face Detection & Feature Extraction	backend.vision
	FR-2.2 (TASK-04)	Centroid Spatial Multi-Participant Tracking	backend.vision
	FR-2.3 (TASK-05)	Sustained Presence & Walk-By Filter Engine	backend.attend
	FR-2.4 (TASK-06)	Head Pose (Yaw/Pitch) & Inactivity Monitoring	backend.vision
EPIC 3	FR-3.1 (TASK-07)	Real-Time Audio PCM Ingestion & Standardization	backend.audio
	FR-3.2 (TASK-08)	Voice Activity Detection (VAD) & Energy Dynamics	backend.audio
	FR-3.3 (TASK-09)	Classroom Acoustic Rhythm & Monologue Detection	backend.audio
EPIC 4	FR-4.1 (TASK-10)	Multimodal Fusion & Temporal Window Alignment	backend.fusion
	FR-4.2 (TASK-11)	Multi-Factor Engagement Index Computation	backend.fusion
	FR-4.3 (TASK-12)	Autonomous Pedagogical Reasoning Agent Core	backend.agent
EPIC 5	FR-5.1 (TASK-13)	Context-Aware Real-Time Teaching Nudges	backend.agent
	FR-5.2 (TASK-14)	Student-Facing Reflective Focus Analytics	backend.core
EPIC 6	FR-6.1 (TASK-15)	Full-Stack FastAPI Gateway & Web HUD Interface	backend.api
	FR-6.2 (TASK-16)	Streamlit Analytics Hub & CSV Data Export	streamlit_app
	FR-6.3 (TASK-17)	91-Test Automated Verification Suite	tests/

Detailed Functional Specifications

The following functional specifications define the operational thresholds, mathematical parameters, and deterministic behavior required across all perception, attendance, and feedback modules.

EPIC 1: Requirements, Configuration & Edge Persistence

- **FR-1.1:** The system shall initialize via a unified `SystemConfig` dataclass supporting configuration via code, environment variables (`.env`), or JSON files.
- **FR-1.2:** The system shall maintain an ACID-compliant local SQLite database (`classroom_intelligence.db`) with tables for `sessions`, `students`, `attendance_logs`, `engagement_metrics`, and `pedagogical_suggestions`. All biometric student IDs shall be hashed using a configurable cryptographic salt.

EPIC 2: Computer Vision & Attendance Engine

• **FR-2.1:** The vision pipeline shall detect face regions in video frames from webcam (`index=0`), simulated feeds, or MP4 files at ***\ge 15*** FPS.

• **FR-2.2:** The multi-participant tracker shall match centroids across successive frames with max distance thresholds, maintaining continuous tracks (`TRK-001`, `TRK-002`) across momentary occlusions.

• **FR-2.3:** The presence engine shall distinguish transient passersby (***<15***s) from enrolled students. Candidate attendance shall require a sustained presence ratio of ***\ge 65\%*** of the observation window.

• **FR-2.4:** Head pose estimation shall compute Yaw and Pitch angles. Yaw deviations ***>25^\circ*** or Pitch deviations ***>20^\circ*** shall classify the student as "Distracted".

EPIC 3: Acoustic Signal Processing & Speech Dynamics

• **FR-3.1:** The audio engine shall ingest raw PCM audio at 16,000 Hz, standardizing multi-channel buffers to single-channel float matrices.

• **FR-3.2:** Energy dynamics shall be calculated via Root Mean Square (RMS) energy. A Voice Activity Detector (VAD) shall flag speech frames using an adaptive threshold (***0.035***).

• **FR-3.3:** The acoustic classifier shall detect continuous instructor monologues exceeding 12 minutes and classroom silence exceeding 6 minutes, emitting diagnostic events. **No Speech-to-Text (STT) or linguistic transcription shall occur.**

EPIC 4: Multimodal Agent & Unified Fusion

• **FR-4.1:** The multimodal fusion engine shall synchronize visual attention frames and acoustic feature chunks into temporal 5-second sliding windows.

• **FR-4.2:** Engagement shall be calculated using the weighted formula:

\text{Engagement Index} = (0.50 \times \text{Visual}) + (0.20 \times \text{Motion}) + (0.30 \times \text{Acoustic})

• **FR-4.3:** The autonomous pedagogical agent shall monitor for engagement score drops ***>25\%*** and trigger contextual intervention rules.

EPIC 5: AI Suggestions & Student-Facing Analytics

• **FR-5.1:** The suggestion engine shall provide pedagogical recommendations (e.g., "Pose a concept check question", "Initiate peer discussion") with a mandatory 45-second cooldown to prevent alert fatigue.

• **FR-5.2:** Student-facing analytics shall compute individual Focus Zone Periods, attentional consistency percentiles, and generate self-improvement insights.

EPIC 6: Interface, Dashboard & System Verification

• **FR-6.1:** The FastAPI gateway shall expose REST endpoints, WebSocket live telemetry feeds (`/ws/telemetry`), and an interactive single-page application (SPA).

• **FR-6.2:** A companion Streamlit dashboard shall display live HUD gauges, historical session heatmaps, focus zone distributions, and export verified attendance records to CSV.

• **FR-6.3:** The system shall pass 100% of the 91 automated tests in `tests/test_system.py`, validating edge cases, memory resilience, and privacy adherence.

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3. Non-Functional Requirements (NFR)

NON-FUNCTIONAL REQUIREMENTS (NFR)			
NFR ID	Category	Requirement Metric	Verification Method
NFR-01	Offline Guarantee	0 bytes external network telemetry	Network packet audit
NFR-02	Privacy-by-Design	0 raw speech-to-text words stored	Codebase grep audit
NFR-03	Vision Latency	Processing latency < 150 ms per frame	Pipeline benchmark
NFR-04	Audio Latency	Processing latency < 500 ms per chunk	Pipeline benchmark
NFR-05	Hardware Agility	Operates on standard CPU (No discrete GPU)	CPU-only runner test
NFR-06	Data Durability	Zero database corruption on unexpected exit	SQLite WAL mode test

Detailed Non-Functional Specifications

1. NFR-01: 100% Edge Offline Operation:

• The system shall run entirely on local host hardware without requiring an active internet connection or external cloud API access. All model weights, cascade files, and databases must be bundled locally.

2. NFR-02: Strict Privacy-by-Design:

- Visual feeds shall support live facial ROI blurring (`blur_kernel_size = 31`).
- Identifiable facial images shall never be stored without explicit instructor linking.
- Microphones shall never be utilized for speech-to-text recording or verbal logging.

3. NFR-03: Low-Latency High-Throughput Execution:

- The end-to-end vision pipeline shall achieve at least 15 FPS on a standard quad-core CPU.
- Total latency between frame capture and WebSocket HUD update shall not exceed 150 milliseconds.

4. NFR-04: Cross-Platform Reproducibility:

- The complete codebase shall execute seamlessly across Windows 10/11, macOS, and Linux without platform-specific binary dependencies.

5. NFR-05: Fault Tolerance & Graceful Degradation:

- If an optional cloud LLM API key (Groq/OpenAI) is missing or invalid, the system shall seamlessly fall back to deterministic rule-based heuristic suggestions with 0 service interruption.
- If the video source drops frames or disconnects, the tracker shall gracefully mark tracks as coasting before removing them.